



LoRa Basics™ Modem-E V2 Reference Manual

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1. Introduction

1.1 Scope

LoRa Basics™ Modem-E is the programmed software embedded inside the LR1110 and LR1121 that supports the LoRaWAN® Standard modem stack, and embeds LoRaWAN® modem and transceiver functions in a single chip.

NOTE: On the LR1110, it is not possible to use GNSS or Wi-Fi functionalities with the latest Modem-E V2 firmware installed.

This document provides information on how to use the LoRa Basics™ Modem-E device in an application, using the features described in [Section 1.4](#).

By default, the Modem-E boots up as a LoRaWAN modem and can receive any modem commands. LoRaWAN Modem and transceiver mode are exclusive, i.e. when LoRaWAN modem is active, a few System commands are also available, but no Radio transceiver commands are available. In order to use the System and Radio transceiver commands, the modem must be suspended with the command `CMD_MODEM_SET_SUSPEND_RADIO_COMMUNICATIONS`. When the modem is suspended it cannot be used, and the host MCU processes the transceiver commands.

The Modem-E can receive file downloads through the LoRaWAN packages ALC-Sync, Remote Multicast Setup and Fragmented Data Block transport.

The Modem-E V2 development board is based on the LR1110 and LR1121 EVK ref design shields, plugged on STM32 Nucleo boards, with 64kB RAM (2x 8kB in retention) and 256kB Flash.

Note: These memory sizes must not be considered as compelling design requirements for the utilization of Modem-E.

Common functionalities of the LoRa Basics™ Modem-E are described in:

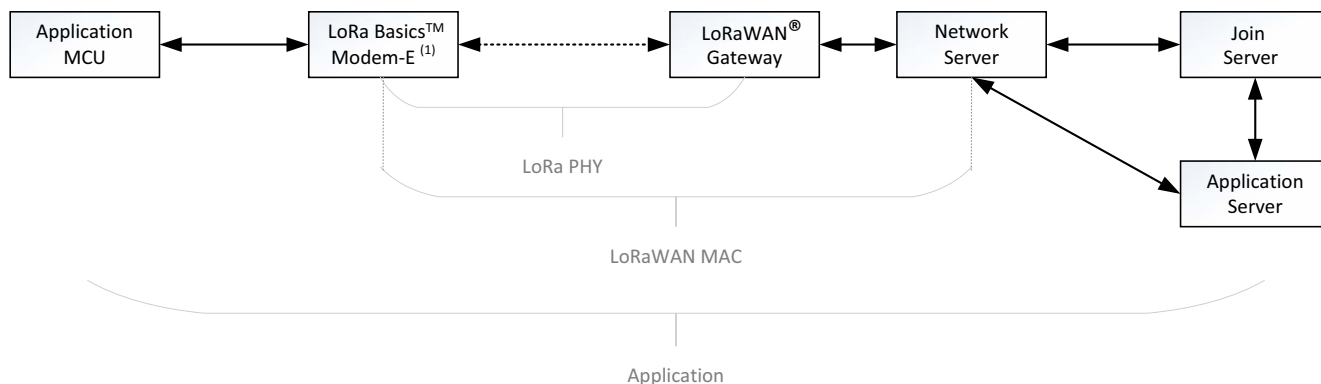
- LR1121 Transceiver User Manual
- LR1121 Data Sheet
- LR1110 Transceiver User Manual
- LR1110 Data Sheet

1.2 Related documents

- | | |
|---------------|--|
| • UM.LR1121 | LR1121 Transceiver User Manual |
| • DS.LR1121 | LR1121 Transceiver Datasheet |
| • UM.LR1110 | LR1110 Transceiver User Manual |
| • DS.LR1110 | LR1110 Transceiver Datasheet |
| • TS001-1.0.4 | LoRaWAN® L2 1.0.4 Specification |
| • TS003-2.0.0 | Application Layer Clock Synchronization |
| • TS004-2.0.0 | Fragmented Data Block Transport |
| • TS005-2.0.0 | Remote Multicast Setup |
| • TS009-1.2.1 | LoRaWAN Certification Protocol Specification |
| • TS011-1.0.0 | Relay |
| • TR013-1.0.0 | Carrier Sense Multiple Access (CSMA) |
| • RP002-1.0.4 | Regional Parameters 2-1.0.4 |

1.3 Overview

LoRa Basics™ Modem-E and LoRaWAN complete system is depicted in Figure 1-1 below.



1.) LoRa Basics Modem-E Firmware running on the LR1110 / LR1121 device

Figure 1-1: LoRa Basics™ Modem-E System Overview

1.4 Implemented Features

The following features are embedded in the LoRa Basics™ Modem-E:

- LoRaWAN® Class A, B and C Standards version 1.0.4 ¹
- Support for Over The Air Activation (OTAA)
- Support for ABP-like activation (Modem-E 2.1.0+) **NOTE:** ABP comes with limitations. Please check the latest drivers and examples.
- Support for configuration save/retrieve in/from internal chip NVM, which is qualified to guarantee up to 10k write cycles.
- Regional parameters:
 - ♦ EU868
 - ♦ US915
 - ♦ AS923_GRP1, AS923_GRP2, AS923_GRP3, AS923_GRP4
 - ♦ AU915
 - ♦ CN470
 - ♦ CN470_RP_1_0
 - ♦ IN865
 - ♦ KR920
 - ♦ RU864
 - ♦ WW2G4 (LR1121 only)
- Multicast class B and C. Up to 2 remote Multicast sessions
- Transceiver commands are supported
- Embedded certification mode
- Relay end-device
- Listen Before Talk (LBT)
- Carrier Sense Multiple Access (CSMA)
- Firmware Update Over the Air (FUOTA) packages
 - ♦ File size limited to 30 kB
 - ♦ One fragmentation session at a time is supported.

1. LoRaWAN® specification v1.0.4 certified, regional parameter 1.0.3

2. Host-Controller Interface

The LoRa Basics™ Modem-E communicates with the host MCU using a set of API commands and responses implemented over SPI. It uses two additional signals to communicate status with the host: BUSY and EVENT.

- **BUSY:** Output, active-low. The BUSY line is high when the Modem is in sleep mode. The HOST MCU must check the BUSY line is high before starting the communication with the Modem.
- **EVENT:** Output, active-high. The same output line is used for Modem Event and transceiver IRQs.

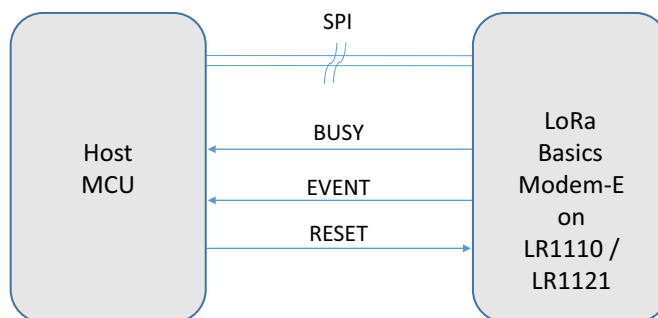


Figure 2-1: LoRa Basics™ Modem-E Interface

All communications consist strictly of host-initiated commands, followed by LoRa Basics™ Modem-E responses. The device does not send any unsolicited response on the SPI port. If some event data becomes available asynchronously, the device signals this using the EVENT signal. The EVENT signal usage is strongly advised for low power applications.

The LoRa Basics™ Modem-E SPI protocol allows the device to return to sleep mode automatically for lowest power consumption. It also features an automatic CRC protection and parameters verification in the SPI commands, so that the host MCU can read a feedback (a.k.a return code) immediately after the command is sent.

In the case of incomplete or aborted commands, the Modem-E has a built-in fail-safe mechanism. In order to be ready to receive new commands, it resets its protocol interface after 2 seconds.

2.1 Host Interrupts

2.1.1 Transceiver Operation (Suspended Mode)

LoRa Basics Modem-E can by-pass the LoRaWAN modem and use the chip as a bare dual band transceiver working with a proprietary protocol running on the external MCU. This transceiver operation is activated through the suspended mode. It's worth noting that the application could dynamically switch between suspended and LoRaWAN modem modes. When the modem is suspended, the EVENT line indicates that an IRQ from a transceiver command is available.

The 32-bit IRQ status can be read by the host through the serial interface. It can be cleared by writing a 1 to the respective bit.

Table 2-1: Interrupt List

Bit	IRQ	Description	Mode	Protocol
2	TxDone	Packet transmission completed	TX	All
3	RxDone	Packet received	RX	All
4	PreambleDetected	Preamble detected	RX	All
5	SyncWordValid / HeaderValid	Valid sync word / LoRa header detected	RX	All
6	HeaderErr	LoRa header CRC error	RX	LoRa
7	Err	Packet received with error LoRa: Wrong CRC received GFSK: get details from GetPacketStatus command	RX	All
8	CadDone	Channel activity detection finished	RX, CAD	LoRa
9	CadDetected	Channel activity detected	RX, CAD	LoRa
10	Timeout	Rx or Tx timeout	RX, TX	All
11	FHSS	IRQ after each ramp-up for intra-packet hopping	TX	LR-FHSS
12	InterPacket1	Host can load new frequencies table	TX	LR-FHSS
13	InterPacket2	Host can load new payload	TX	LR-FHSS
14	RngReqVld	Valid ranging request received (slave)	RX	Ranging
15	RngReqDis	Ranging request discarded (no address match)	RX	Ranging
16	RngRespDone	Slave sent its ranging response	RX	Ranging
17	RngExchVld	Master received a valid ranging response from the slave	TX	Ranging
18	RngTimeout	Master did not receive a response from the slave	TX	Ranging
21	EOL	End of life	All	
22	Cmd Error	There was a host command fail/error	All	
23	Error	An error other than a command error occurred (See GetErrors)	All	
24	FskLenError	IRQ raised if the packet was received with a length error	RX	FSK
25	FskAddrError	IRQ raised if the packet was received with an address error	RX	FSK
27	LoRaRxTimestamp	Last LoRa symbol received. To be used for time stamping the received packet. Device is still in Rx mode	RX	

2.1.2 Modem Mode

In Modem mode, when the modem is not suspended, the EVENT line signals to the host that the modem has asynchronous event data pending from the modem or LoRaWAN stack.

- The host MCU must use the ModemGetEvent command to retrieve such data.
- All transceiver interrupts are disabled and cleared when the modem is activated.

The host MCU must clear all events before returning to sleep mode. Otherwise, there are two possible consequences:

- First, the MCU host might miss the rising edge of the EVENT line when there is a new event.
- Second, when the EVENT line is high, the Modem-E consumes several tens of μA , which is not good for low power design.

2.2 SPI Commands

An SPI transaction is initiated by a falling edge of the NSS signal.

Upon generation of an SPI command, the LoRa Basics™ Modem-E analyses the command message (correct number of bytes and correct CRC), interprets the command, and finally returns a response.

- If the packet structure is invalid or the CRC wrong, the LoRa Basics™ Modem-E ignores the packet and returns the FrameError response code. A response is returned to most commands (but not Reset and Reboot).
- In all SPI transactions, multiple octet fields are sent MSB first.
- Negative values transmitted over the SPI are expressed as two's complement, unless otherwise indicated.

The Modem-E can receive System, Radio and Modem commands (System and Radio commands are legacy transceiver commands). The SPI protocol is the same for all commands.

The return code indicates the status of the command.

For transceiver commands (System and Radio) only 5 return codes are possible:

- OK: The command is validated and executed.
- Unknown: The command is not available.
- Invalid: Parameters of the command are invalid or the command is launched without suspending the modem. The list of transceiver commands where the suspension of the modem is mandatory is defined below.
- Fail: The command failed.
- FrameError: SPI checksum error.

Radio commands are available only if the modem is suspended. For System commands see below for each API function if the suspension of the modem is necessary. If the modem is suspended transceiver IRQs are available.

The maximum SPI baud rate is 16Mbps. In all transactions on the SPI, multiple octet fields use the big endian format.

2.2.1 Command Format

The SPI command consists of a 2 byte Opcode (Group Id + Cmd), followed by a Payload of variable length (N), and then a CRC, as shown in [Table 2-2: SPI Command Structure](#) below.

Table 2-2: SPI Command Structure

Byte	0	1	2:N+1	N+2
Field	Group Id	Cmd	Payload	CRC

- The Group Id is defined as indicated in [Table 2-3: Command Group Id Definition](#).

Table 2-3: Command Group Id Definition

Group Id	Group
0x01	System
0x02	RadioCtrl
0x06	Modem

- *Cmd*: Command code
- *Payload*: Binary data parameters
- *CRC*: The 8-bit CRC is computed over the whole SPI command packet, using a polynomial generator 0x65 (reversed reciprocal), and a CRC initial value of 0xFF. For example $\text{crc8}('123456789') = 0x18$.

2.2.2 Response Format

The SPI response consists of a 1-byte Return Code, followed by an optional N byte(s) Payload (binary data), and then a CRC, as shown in [Table 2-4: SPI Response Structure](#) below:

Table 2-4: SPI Response Structure

Byte	0	1:N	N+1
Field	Return Code	Payload (Optional)	CRC

The Return Codes (also referred as RC) are defined in [Table 2-5: SPI Response Return Code Definitions](#).

Table 2-5: SPI Response Return Code Definitions

Value	Return Code (RC)	Description	Command Group Id
0x00	OK	Command executed without error	System/RadioCtrl/Modem
0x01	Unknown	Command code unknown	System/RadioCtrl/Modem
0x02	NotImpl	Command not yet implemented	Modem
0x03	NotInit	Command not initialized	Modem
0x04	Invalid	Command parameters invalid	System/RadioCtrl/Modem
0x05	Busy	Command cannot be executed now	Modem
0x06	Fail	Command execution failed	System/RadioCtrl/Modem
0x08	BadCrc	CRC check failed	Modem
0x0A	BadSize	Size check failed	Modem
0x0F	FrameError	SPI checksum / CRC error	System/RadioCtrl/Modem
0x12	NoEvent	No event available	Modem

2.2.3 SPI Write Transactions

An SPI Write consists of two SPI transactions. The host MCU must ensure that the BUSY signal is high before sending the command. The SPI write transaction is depicted in [Figure 2-2: SPI Write Transaction](#) hereafter.

- Before the host issues the SPI command, the LoRa Basics™ Modem-E is in sleep mode, with BUSY signal set to high.
- The host MCU toggles the NSS line to wake up the LoRa Basics™ Modem-E, and waits for the BUSY signal to go low.
- Once BUSY signal is low, the host MCU drives the NSS line low and sends the bytes stream.
- After the bytes stream is sent, the host MCU drives the NSS line high, and waits for the BUSY signal to go high (indicating that LoRa Basics™ Modem-E is processing commands).
- Once the BUSY signal is high, the host MCU drives the NSS line to low again and sends two NOPs (0x00 bytes).
- After sending two NOPs (0x00 bytes), the host MCU drives the NSS line high, and waits for the BUSY signal to go low.
- The LoRa Basics™ Modem-E drives the BUSY signal low when it is processing the command.
- If no more processing is needed, the LoRa Basics™ Modem-E goes to sleep mode and drives the BUSY signal high.

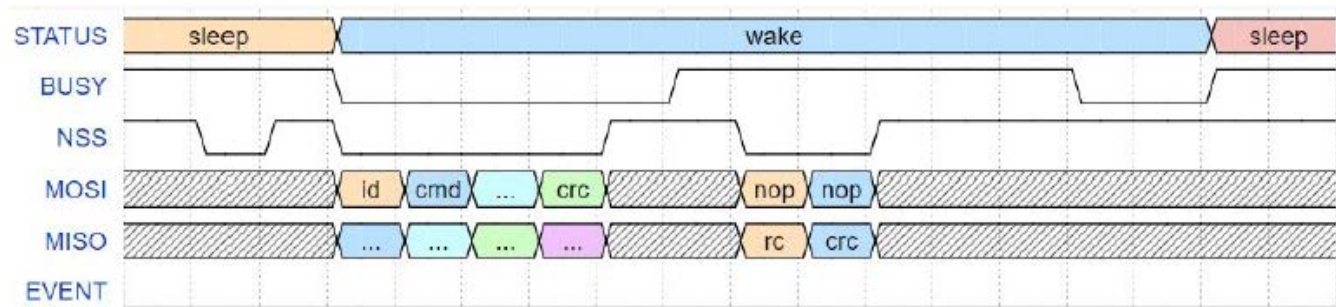


Figure 2-2: SPI Write Transaction

NOTE: No information is sent back to the host MCU on the MISO when the host MCU sends id + cmd to the Modem

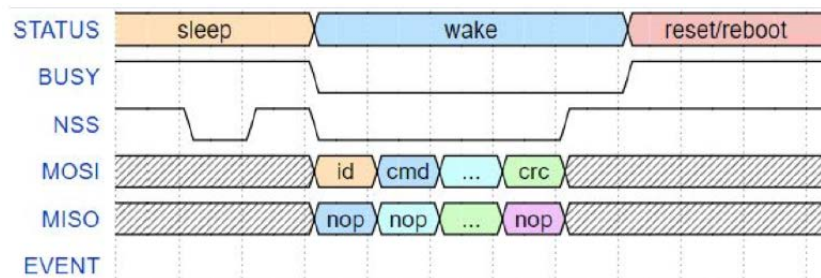


Figure 2-3: SPI Write Command Without Return Codes

There are several commands of exception because they do not generate return codes, such as:

- System command: *Reboot(...)*
- Modem commands: *FactoryReset/Reset(...)*
- Modem test commands (when the modem is in test mode): *TST_BUSYLOOP*, *TST_PANIC*, *TST_WATCHDOG*.
If the modem is not in test mode, all test mode commands are rejected.

2.2.4 SPI Read Transactions

An SPI read consists of two SPI transactions. The host MCU must ensure that the BUSY signal is high before initiating the SPI transactions. The SPI read transaction is depicted in [Figure 2-4: SPI Read Transaction](#) hereafter.

1. Before the host issues the SPI command, the LoRa Basics™ Modem-E is in sleep mode, with the BUSY signal set to high.
2. The host MCU toggles the NSS line to wake up the LoRa Basics™ Modem-E and waits for the BUSY signal to go low (less than 1ms typ).
3. Once BUSY signal is low, the host MCU drives the NSS line low and sends the bytes stream.
4. After the bytes stream is sent, the host MCU drives the NSS line high, and waits for the BUSY signal to go high.
5. Once BUSY signal is high, the host MCU drives NSS line low again and sends an expected number of NOPs (0x00 bytes).
6. After sending the NOPs (0x00 bytes), the host MCU drives the NSS line high, and waits for the BUSY signal to go low.
7. The LoRa Basics™ Modem-E drives the BUSY signal low while it is processing.
8. If no more processing is needed, the LoRa Basics™ Modem-E goes to sleep mode and drives the BUSY signal high.

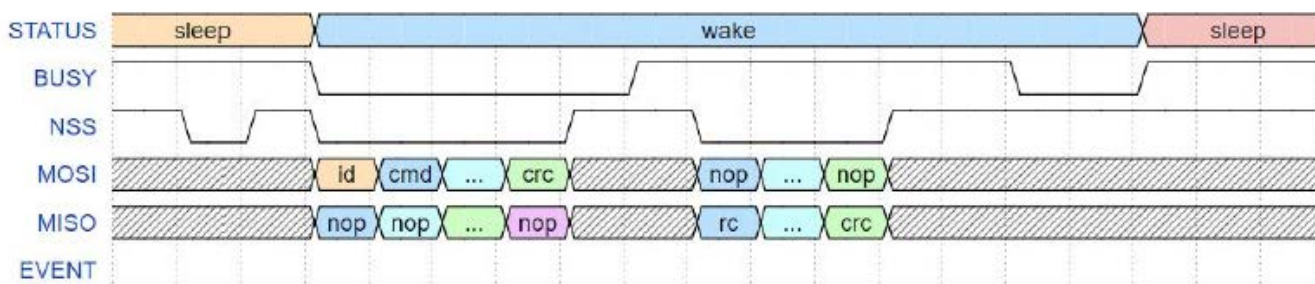


Figure 2-4: SPI Read Transaction

NOTE: No information is sent back to the host MCU on the MISO when the host MCU sends id + cmd to the Modem.

For the case where the return code is not 0, no payload data exists between the return code and CRC. The host MCU should recognize the byte of RC to locate the CRC byte systematically.

2.2.5 Modem Command Format

Each modem command must start with Opcode 0x06, SubgroupID then CommandID before the first byte of the requested command.

Table 2-6: Modem Request Command Format

Byte	0	1	2	[3:N-1]	N
Field	CMD_MODEM_GROUP (0x06)	SubgroupID	CommandID	Command payload	CRC

Table 2-7: Modem SubgroupID definition

Value	Details
0	BSP commands
1	Modem commands
2	LoRaWAN commands
3	LoRaWAN Relay commands

Table 2-8: Modem Response Command Format

Byte	0	[1:N-1]	N
Field	Return Code	data	CRC

2.2.6 Transceiver Command Format

Transceiver commands represent Register/Memory access operations, System configuration and status operations, and Radio configuration and status operations.

Table 2-9: Transceiver Request Command Format

Byte	0	1	[2:N-1]	N
Field	CMD_TRANSCEIVER_GROUP_ID (0x06)	CommandID	Command payload	CRC

Table 2-10: Transceiver SubgroupID definition

Value	Details
1	Register/Memory access and System commands
2	Radio Control commands

Table 2-11: Transceiver Response Command Format

Byte	0	[1:N-1]	N
Field	Return Code	data	CRC

2.3 EVENT Signaling

The EVENT signal is an output signal of the LoRa Basics™ Modem-E. It is mapped on the DIO9 pin of the LoRa Basics™ Modem-E.

This line signals to the host MCU that the device has asynchronous event data pending. The event can be LoRaWAN® stack or RadioCtrl events. The host MCU must use the *ModemGetEvent (...)* command to retrieve such data.

The EVENT signal stays high until all events are cleared. The host MCU must clear all the events before returning to sleep mode. This prevents the host MCU from missing the rising edge of the EVENT signal when there is a new event. Secondly, this saves several tens of uA of current consumption when the EVENT signal is kept low by the LoRa Basics™ Modem-E.

2.3.1 Event Handling Examples

Some SPI commands generate an immediate response by the LoRa Basics™ Modem-E, and therefore do not generate any event, such as the command *RequestTx(...)*, as shown in [Figure 2-5: RequestTx Transaction](#).

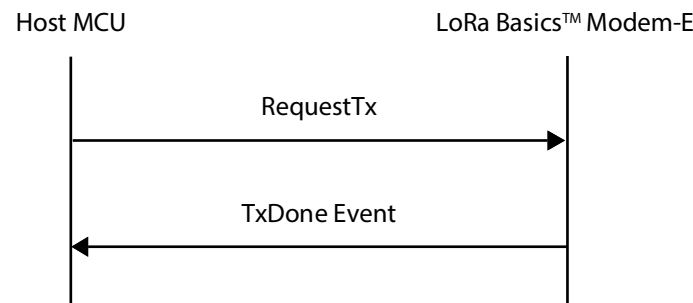


Figure 2-5: RequestTx Transaction

[Figure 2-6: Class A RequestTx Transaction](#) (page 17) shows the event sequence of a LoRaWAN® Class A packet transmission.

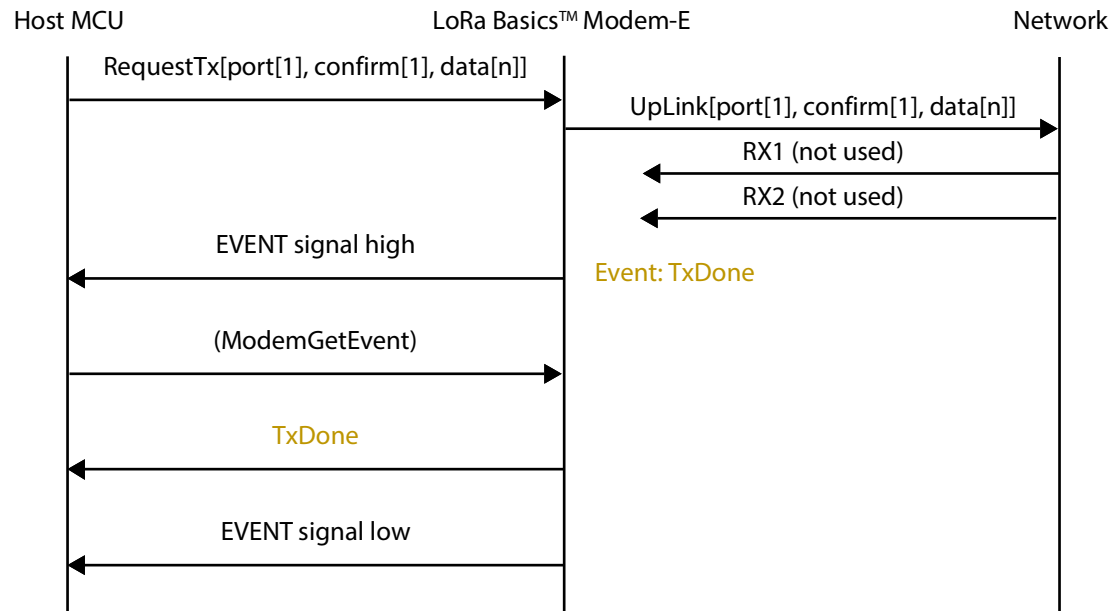


Figure 2-6: Class A RequestTx Transaction

2.3.2 ModemGetEvent Command

The command *ModemGetEvent(...)* retrieves pending events from the Modem. Pending events are indicated by the EVENT signal. The EVENT signal is de-asserted after all events have been retrieved and no further events are available. When no event is available this command returns NoEvent (0x12).

All events that are not retrieved by the application, except DownData, will be overwritten by new event data of the same type. In this case, only the latest event data will be returned and the count field indicates how many events of this type have been missed. DownData is the only event stored in a 512 byte FIFO buffer, it is advised to fetch it as soon as possible in order to avoid losing payloads. The most recent event is returned first by the *ModemGetEvent(...)* command.

This command returns Modem Event information. When a modem service-related event occurs, the modem drives the Event line high to indicate the presence of events to the Host MCU. The ModemEvent is always 4 bytes. If the RC is NoEvent, the event returns no data. The user should send Get Event commands until they receive a NoEvent to clear the event buffer.

Table 2-12: ModemGetEvent Command

Byte	0	1	2	2
Field	0x06	0x01	0x04	CRC

- Byte 0: 0x06 indicates that this command is from the Modem group
- Byte 1: 0x01 indicates that this command is in the Modem sub-group
- Byte 2: 0x04 indicates that this command is the ModemGetEvent command

Table 2-13: ModemGetEvent Response

Byte	0	1	2	3:4	5
Data to Host	ReturnCode	EventType	MissedEventsCount	EventData	CRC

- *ReturnCode* (RC) is defined in [Table 2-5: SPI Response Return Code Definitions](#). if the RC is NoEvent, the payload size is still 4 bytes but the payload should be ignored.
- ModemEvent:
 - ♦ Byte 1: *EventType*: Defined in [Table 2-14: ModemEvent: EventType and EventData](#).
 - ♦ Byte 2: *MissedEventsCount*: Indicates the number of events of this type missed.
 - ♦ Byte 3-4: *EventData*: Payload.

Table 2-14: ModemEvent: EventType and EventData (Sheet 1 of 3)

Event Type	Code	Description	Event Data
Reset	0x00	Modem has been reset. See Section 3.5 for reset event triggers.	RFU[2]
Alarm	0x01	Programmed alarm timer has expired	RFU[2]
Joined	0x02	Modem has successfully joined Network	RFU[2]
JoinFail	0x03	Attempt to join network failed. A JoinFail event is generate each time a JoinRequest fails. The JoinFail event can be generated repeatedly, the join process continues until the network is joined or is aborted by the LeaveNetwork command.	RFU[2]

Table 2-14: ModemEvent: EventType and EventData (Sheet 2 of 3)

Event Type	Code	Description	Event Data
TxDone	0x04	RequestTx or EmergencyTx operation has completed. Raised after a RequestUplink, EmergencyUplink or EmptyUplink command: 0x00: NOT_SENT: Payload was not sent over the air by the modem 0x01: SENT: Payload was sent, no ACK received from network server 0x02: CONFIRMED: Payload sent, ACK received from network server	Status[1] RFU[1]
DownData	0x05	A downlink with application data has been received. Command GETDOWNLINK_DATA_SIZE must be called before the GETDATA and GETDOWNLINK_METADATA. The size indicates the number of bytes that must be read over the SPI to get the downlink data.	RFU[2]
LinkCheck	0x06	Generated after a LoRaWanMacRequest with bit LinkCheckReq. Gives the status of the request and data received from the network server: 0x00: LINK_CHECK_NOT_RECEIVED: Link check response not received 0x01: LINK_CHECK_RECEIVED: Link check response has been received	Status[1] RFU[1]
LORAWAN_MAC_TIME	0x07	Occurs when the time sync status of LoRaWAN mac device time changes. Status byte indicates if the time is valid or not: 0x00: TIME_NOT_VALID: Time is not valid 0x01: TIME_VALID: Time is valid and has been synchronized 0x02: TIME_VALID_BUT_NOT_SYNC: Time is valid but not synchronized	Status[1] RFU[1]
CLASS_B_PING_SLOT_INFO	0x08	Generated after a ping slot info request to the Network Server. Status shows if the Network Server has answered (i.e. received it) or not: 0x00: PING_SLOT_NOT_RECEIVED: ping slot info request not answered 0x01: PING_SLOT_RECEIVED: ping slot info request answered	Status[1] RFU[1]
CLASS_B_STATUS	0x09	Generated if class B is ready to receive downlinks (i.e. the status of class B has been changed internally). Status indicates if Class B is ready to receive downlinks, or not: 0x00: CLASS_B_NOT_READY: Class B is not ready to receive downlinks 0x01: CLASS_B_READY: Class B is ready to receive downlinks	Status[1] RFU[1]
NEW_MULTICAST_SESSION_CLASS_C	0x0A	Generated when a new multicast session in class C starts. 0x00: NO_CLASS_C_SESSION: Multicast class C session has not started 0x01: CLASS_C_SESSION_STARTING: Multicast class C session is starting	group_ID[1] RFU[1]
NEW_MULTICAST_SESSION_CLASS_B	0x0B	Generated when a new multicast session in class B starts. 0x00: NO_CLASS_B_SESSION: Multicast class B session has not started 0x01: CLASS_B_SESSION_STARTING: Multicast class B session is starting	group_ID[1] RFU[1]
NO_MORE_MULTICAST_SESSION_CLASS_C	0x0C	Generated when the multicast session in class C ends.	RFU[2]
NO_MORE_MULTICAST_SESSION_CLASS_B	0x0D	Generated when the multicast session in class B ends.	RFU[2]
RELAY_TX_DYNAMIC	0x0E	Relay TX dynamic mode has enabled or disabled the WOR protocol. Status indicates if relay Tx dynamic mode has enabled or disabled the WOR protocol. 0x00: WOR_PROTOCOL_DISABLED: WOR protocol has been disabled 0x01: WOR_PROTOCOL_ENABLED: WOR protocol has been enabled	Status[1] RFU[1]

Table 2-14: ModemEvent: EventType and EventData (Sheet 3 of 3)

Event Type	Code	Description	Event Data
RELAY_TX_MODE	0x0F	Relay TX activation has been updated. Status indicates if the relay Tx dynamic mode has enabled or disabled the WOR protocol. 0x00: RELAY_TX_DISABLED: Relay Tx is disabled 0x01: RELAY_TX_ENABLED: Relay Tx is enabled 0x02: RELAY_TX_DYNAMIC: Relay Tx is dynamic 0x03: RELAY_TX_END_CONTROL: Relay Tx is end device controlled	StatusMode[1] RFU[1]
RELAY_TX_SYNC	0x10	Relay TX synchronization has changed. Status indicates if the relay Tx dynamic mode has enabled or disabled the WOR protocol. 0x00: RELAY_TX_SYNC_INIT: Relay TX has been initialized 0x01: RELAY_TX_UNSYNC: Relay Tx is not synchronized 0x02: RELAY_TX_SYNC: Relay Tx is synchronized	Status[1] RFU[1]
ALCSYNC_TIME	0x11	Time is synchronized	RFU[2]
FUOTA_DONE	0x12	This event indicates the status of the FUOTA (firmware update over the air) with fragmented data block service. 0x00: FUOTA_DONE: FUOTA finished successfully 0x01: FUOTA_FAIL: FUOTA failed -1: FUOTA_ONGOING: FUOTA is ongoing -2: FUOTA_NOT_STARTED: FUOTA has not started	Status[1] RFU[1]
TEST_MODE	0x13	This event indicates the status of the Test Tx mode. 0x00: TEST_MODE_NOT_SENT: Test Tx mode not sent 0x01: TEST_MODE_SENT: Test Tx mode sent 0x02: TEST_MODE_END: Test Tx mode ended action	Status[1] RFU[1]
NO_DOWNLINK_THRESHOLD	0x15	ADR back-off reaches its end or when user-defined no RX threshold is reached BitMask[1]: - bit0: ADR_BACKOFF_END - bit1: NO_RX_USER_THRESHOLD	BitMask[1] RFU[1]
RESET_REQUEST	0x16	The modem needs to be reset by an internal service, for example the LoRaWAN certification could request a DUT reset. On this event the host can backup the modem context then reset the device.	RFU[2]

2.4 Sleep Mode

In sleep mode, the watchdog is not stopped and the DIOs are in Hi-Z mode.

3. Transceiver Interface

Please refer to the Transceiver User Manual for detailed information, and for a list of the commands, see [Section 5](#).

3.1 Register/Memory Commands

3.1.1 RF Switch

The RF Switch must be powered by the Modem-E, and not always powered-on to avoid wasting a lot of power.

The RF Switch must not be controlled by the MCU as the Modem-E can send uplinks without the MCU being notified. You can see how this situation is handled on the EVK available here:

https://raw.githubusercontent.com/LoRa-net/lora_edge_tracker_ref_design/master/smtc_tracker_app/LR1121_e595v03a_schematic.pdf, on page 2 section B4, for chip U2 (PE4259 RF switch). Note that VDD and CTRL pins of U2 are both controlled by the Modem-E on DIO5 and DIO6.

The other commands are identical (same Opcode, same arguments, same functionality).

3.1.2 LF Clock Configuration

By default, the Modem-E V2 boots with the internal RC32kHz as the LF Clock, and the flag Wait for Ready is True.

3.2 Radio Commands

Radio commands are accepted if the modem is suspended with the modem command *ModemSetSuspendRadioCommunications()*.

Some test commands allow an evaluation of the LoRa Basics™ Modem-E radio parameters. See the Modem-E Driver documentation for the radio test command description.

3.3 System Modes

The LoRa Basics™ Modem-E uses the same system modes as the transceiver firmware but ensures automatic transitions between the different modes and sleep mode.

By default, the LoRa Basics™ Modem-E is in sleep mode for lowest power consumption. Toggling the NSS pin wakes-up the LoRa Basics™ Modem-E, so that it is ready to accept any command, or send a response byte stream (refer to [Section 2.2.3 "SPI Write Transactions" on page 13](#), and [Section 2.2.4 "SPI Read Transactions" on page 14](#)).

At the end of the command processing, the LoRa Basics™ Modem-E automatically returns to sleep mode.

3.4 System Commands

Cryptographic Engine commands are not supported, but all other commands that are supported by the Transceiver firmware are supported by the LoRa Basics™ Modem-E, using the same opcode, same arguments, and same functionality.

Note 1: Exercise caution when configuring Pin DIO11.

It can be configured as both input for XTAL 32kHz (*ConfigLFClock(...)*) and output (*SetDioAsRfSwitch(...)*).

These two use cases are incompatible, therefore care must be taken to not configure the LF clock with Crystal Oscillator and enable the DIO11 pin as an RF Switch at the same time.

Note 2: DCDC to LDO selection is done manually by the user, and depends on the application hardware configuration (electrical schematic). The DCDC configuration is set by default.

3.5 Reset Sources

In the LoRa Basics™ Modem-E, there are multiple ways to trigger a software reset:

- ♦ SPI API modem command *Reset(...)*
- ♦ Watchdog time-out

The command *Reboot(...)* has also the effect of triggering a reset, but this function is meant for firmware updates. For a description of this command, refer to the Transceiver User Manual.

Each reset source generates a *Reset* event. The host MCU must wait for this *Reset* event before sending any command.

The following parameters are stored in NVM, which is qualified to guarantee up to 10K write cycles, and are therefore kept after a software reset:

- ♦ LoRaWAN Certification enabled status
- ♦ DevEui. Default value: as per device provisioning in Semtech production flow
- ♦ JoinEui. Default value: as per device provisioning in Semtech production flow
- ♦ Region. Default value = EU868
- ♦ DevNonce and joinNonce

3.6 Flash Memory

The LR1110 and LR1121 devices embed a Flash memory which stores the internal firmware, the application configuration data and the cryptographic keys. When using a LoRa Basics™ Modem-E, one needs to pay attention to internal flash memory usage to avoid premature aging (flash memory endurance is available in the transceiver datasheet). Several parameters are saved in the non-volatile memory, and are updated only if the value changes compared to what is already stored:

- DevNonce and joinNonce (stored as soon as the "join request" packet is sent)
- DevEUI
- JoinEUI
- NwkKey
- Region
- LoRaWAN Certification enabled status

While DevEUI / JoinEUI / NwkKey / Region parameters are not expected to be modified often, DevNonce & reset counter evolution can become problematic if a join procedure is initiated while having no LoRaWAN coverage.

In LoRa Basics™ Modem-E, LoRaWAN join procedure is implemented according to table 56 "Transmit duty-cycle limitation" from the LoRaWAN Link Layer Specification v1.0.4. This implementation means that the flash memory is preserved in case of no coverage. Nevertheless, the sequence is restarted from scratch if LoRa Basics™ Modem-E is reset - for instance, trying to keep the fastest pace by resetting LoRa Basics™ Modem-E on a regular basis might lead to a premature aging of the flash memory.

3.7 System Boot

At startup, the Modem-E boot uses the internal 32.768kHz crystal oscillator.

The user can change the LfClock source with the *ConfigLfClock(...)* command.

4. Command Sequences

This section provides guidelines on how to configure the LoRa Basics™ Modem-E in order to fulfil some specific functions.

4.1 Device Configuration After Reset

After a reset, the LoRa Basics™ Modem-E device requires the *Join(...)* command to rejoin the network.

Before sending the *Join(...)* command, the host MCU must check and reconfigure the following fields:

1. Configure the application hardware:
 - ♦ Regulator mode (DC-DC or LDO), using the command *SetRegMode(...)*.
 - ♦ TCXO, with the command *SetTcxoMode(...)*.
 - ♦ *RegTcxoTune* = 0 (1.6 V supply voltage).
 - ♦ *Delay* = 255 * 30.52 us (7.7 ms timeout).
 - ♦ PA configuration, through the command *BSPSetRFOutput(...)*.
 - ♦ RF switch control, using the command *SetDioAsRfSwitch(...)*. See [Section 3.1](#) for more precision.
 - ♦ LfClk, using the command *ConfigLfClock(...)*.
2. Configure the Tx power offset: *BSPSetTxPowerOffset(...)*.
3. Configure the Alarm timer if necessary: *SetAlarmTimer(...)*.
4. Configure the LoRaWAN® Class: *SetClass(...)*.
5. Configure the LoRaWAN® Region: *SetRegion(...)*.

The host can save the modem's internal context to NVM. After reboot, the host can restore it, allowing the modem to resume its LoRaWAN session without re-joining.

4.2 LoRaWAN® Join and Packets Transmission

The context can be stored and retrieved if the chip either goes into sleep mode without retention, or enters a POR condition. Both conditions may occur in super energy efficiency applications, for instance in the case of Energy Harvesting where power cannot be maintained all the time.

1. Store context before sleep without retention with `modem_e_store_state_snapshot_to_nvm`
2. Restore after reboot, before any other command, send the restore command with a payload containing the elapsed time in milliseconds since the backup with API `modem_e_restore_state_snapshot_from_nvm`

For an example on how to use the context store/retrieve APIs, please check the certification-abp example in repository https://github.com/Lora-net/ModemE_application_examples.

On success, the modem resumes normal operation with its full LoRaWAN context (session keys, frame counters, MAC state, DeviceTimeReq).

4.2.1 Over-The-Air Activation (OTAA)

1. Get Device EUI using `GetDevEui(...)` command.
2. Register Device EUI on the Network Server.
3. Get Join EUI and Application Key from the Network server.
4. Set the Join EUI using `SetJoinEui(...)` command.
5. Set the APP Key using `SetAppKey(...)` command.

After configuring the keys:

1. Configure the LoRaWAN® parameters:
 - ♦ Set the class with the `SetClass(...)` command.
 - ♦ Set the region with the `SetRegion(...)` command.
2. Start joining with the `Join(...)` command (if using OTAA).
3. At this point the Host MCU waits for the event JOIN_ACCEPT (if using OTAA).
4. Set the ADR profile using the `SetAdrProfile(...)` command.
5. Send packets using the `RequestTx(...)` command.
6. Wait for the event `TxDone`.

NOTE: Semtech Lora Cloud Modem, Geolocation server and Semtech Join Server have been discontinued and were phased out on July 31st 2025. Read here the details about the existing alternatives from partners: <https://www.semtech.com/cloud-geolocation>.

4.2.2 Activation By Personalization (ABP)

In ABP, the device address, network, and application private keys must already be stored and match in the device and server.

There is no joining concept: an ABP session can be started by calling the appropriate driver function and passing the keys and device address.

NOTE: ABP has some limitations. For more information, please see the certification-abp example in repository https://github.com/Lora-net/ModemE_application_examples.

4.3 Use An External Low Frequency Clock

The steps to use an external low frequency (LF) clock are the following:

1. Configure the low frequency clock as RC (0x00) using `ConfigLFClock(...)` command.
2. Configure the low frequency clock as External (0x02) using `ConfigLFClock(...)` command.

4.4 Listen Before Talk

Listen Before Talk (LBT) **must** be used for Japan and Korea regions, and is not necessary in other region. If polite spectrum access is desired in other regions, the use of CSMA is advised.

5. List Of Transceiver Operations

5.1 List of Register/Memory Access Commands

See the Transceiver User Manual for details of these commands.

Table 5-1: Register/Memory Access Commands

Command	Cmd Id	Description	Input	Output	Needs Modem Suspended
WriteAuxReg32	0x0103	Writes data at given aux address. Address must be 32-bit aligned	Addr(31:0) Data(31:0)		No
ReadAuxReg32	0x0104	Reads data at given aux address. Address must be 32-bit aligned and data length in word(32-bit) < 65	Addr(31:0) Len(7:0)	Data	No
WriteRegMem32	0x0105	Writes data at given register/memory address. Address must be 32-bit aligned and data length must be a multiple of 4	Addr(31:0) Data[1..256]		No
ReadRegMem32	0x0106	Reads data at given register/memory address. Address must be 32-bit aligned and data length in word(32bit) < 65	Addr(31:0) Len(7:0)	Data	No
WriteBuffer8	0x0109	Writes data to radio tx buffer	Data[1..256]		No
ReadBuffer8	0x010A	Reads data from radio rx buffer	Offset(7:0) Len(7:0)	Data	No
ClearRxBuffer	0x010B	Clears all data from the radio rx buffer			No
WriteRegMemMask32	0x010C	Read-Modify-Writes data at given register/memory address. Address must be 32-bit aligned	Addr(31:0), Mask(31:0) Data(31:0)		No

5.2 List of System Configuration / Status Commands

See the Transceiver User Manual for details of these commands.

Table 5-2: System Configuration / Status Commands (Sheet 1 of 2)

Command	Cmd Id	Description	Input	Output	Needs Modem Suspended
GetStatus	0x0100	Returns status of device		Status variables	YES
GetVersion	0x0101	Returns version of firmware		Version	NO
GetErrors	0x010D	Returns error status of the device		ErrorStat(15:8) ErrorStat(7:0)	YES
ClearErrors	0x010E	Clears error bits in error status			YES
Calibrate	0x010F	Calibrates the requested blocks according to parameter	CalibParams(7:0)		YES
SetRegMode	0x0110	Sets if BUCK may be enabled for XOSC, FS, RX or TX mode	RegMode (0 = LDO, 1 = BUCK)		NO
CalibImage	0x0111	Launches an image calibration. Freq1, Freq2, in 4MHz steps	Freq1(7:0) Freq2(7:0)		YES
SetDioAsRfSwitch	0x0112	Sets up the RFSWx output configurations for each radio mode	RfswEnable(7:0), RfswStbyCfg(7:0), RfswRxCfg(7:0), RfswTxCfg(7:0), RfswTxHPCfg(7:0), RfswTxHFCfg(7:0)		NO
SetDioIrqParams	0x0113	Configures IRQs to output on IRQ pin(s)	IrqToEn(31:0), IrqToEn2(31:0)		NO
ClearIrq	0x0114	Clears pending irq	IrqsToClear(31:0)		YES
ConfigLFClock	0x0116	Configures the used LF clock	LfClockSetup (7:0)		NO
SetTcxoMode	0x0117	Configures the chip for a connected TCXO	Tune(7:0) Timeout(24:0)		NO
Reboot	0x0118	Reboots (SW reset) the device	StayInBootloader		NO
GetVbat	0x0119	Gets Vbat		VBat(7:0)	NO
GetTemp	0x011A	Gets temperature		Temp(15:8) Temp(7:0)	NO
SetSleep	0x011B	Sets chip into SLEEP mode	SleepConfig SleepTime(31:0)		YES
SetStandby	0x011C	Sets chip into RC or XOSC mode	StdbYConfig (0 = RC, 1 = XOSC)		YES

Table 5-2: System Configuration / Status Commands (Sheet 2 of 2)

Command	Cmd Id	Description	Input	Output	Needs Modem Suspended
SetFs	0x011D	Sets chip into FS mode			YES
GetRandomNumber	0x0120	Gets a 32-bit random number NOTE: This is not a TRNG.		Rnd(31:24) Rnd(23:16) Rnd(15:8) Rnd (7:0)	NO
EraseInfoPage	0x0121	Erase the data in the given flash info page	InfoPage		YES
WriteInfoPage	0x0122	Writes data in the given flash info page, at the given word address. Data length must be a multiple of 4.	InfoPage, In-foWordAddr(15:0), Data[1..256]		YES
ReadInfoPage	0x0123	Reads data in the given flash info page, at the given word address. Data length in word(32bit) < 65.	InfoPage, In-foWordAddr(15:0), InfoReadLen	Data	YES
GetChipEui	0x0125	Returns the 8-byte the factory UID = ChipEui (big-endian)		UID0-7	NO
GetJoinEui	0x0126	Returns the 8-byte factory JoinEUI (big-endian)		EUI0-7	NO
EnableSpiCrc	0x0128	Enables / disables an 8-bit CRC on the SPI interface	Enable, CRC		YES
DriveDiosInSleepMode	0x012A	Enables/disables pull ups/downs on configured RF switch and IRQ dios for sleep modes	Enable		YES

5.3 List of Radio Configuration / Status Commands

See the Transceiver User Manual for details of these commands.

All these commands require that the modem is suspended.

Table 5-3: Radio Configuration / Status Commands (Sheet 1 of 3)

Commands	Cmd Id	Description	Input	Output
ResetStats	0x0200	Resets Rx statistics		NbPkt Received(15:8) NbPkt Received(7:0) NbPkt CrcError(15:8) NbPkt CrcError(7:0)
GetStats	0x0201	Gets Rx statistics		Data1(15:8) Data1(7:0) Data2(15:8) Data2(7:0)
GetPacketType	0x0202	Gets current radio protocol		PktType
GetRxBufferStatus	0x0203	Gets Rx buffer status		Payload length RX RxStart BufferPointer
GetPacketStatus	0x0204	Gets Rx packet status		GFSK: RxStatus (31:24) RxStatus (23:16) RxStatus (15:8) RxStatus (7:0) LoRa: RssiPkt SnrPkt SignalRssiPkt
GetRssiInst	0x0205	Gets instantaneous RSSI		RSSI
SetGfskSyncWord	0x0206	Set the 64 bit GFSK syncword	Syncword	
SetLoRaPublicNetwork	0x0208	Changes LoRa sync work for private or public network	PublicNetwork	
SetRx	0x0209	Set chip into RX mode	Timeout(23:0)	
SetTx	0x020A	Set chip into TX mode	Timeout(23:0)	
SetRfFrequency	0x020B	Set PLL frequency	RfFreq (31:0)	
AutoTxRx	0x020C	Activate or deactivates the auto Tx auto Rx mode	Delay(23:0) IntermediaryMode(7:0) Timeout2(23:0)	

Table 5-3: Radio Configuration / Status Commands (Sheet 2 of 3)

Commands	Cmd Id	Description	Input	Output
SetCadParams	0x020D	Configure LoRa Cad mode	SymbolNum(7:0) DetPeak(7:0) DetMin(7:0) ExitMode(7:0) Timeout(23:0)	
SetPacketType	0x020E	Define radio protocol (FSK, LoRa, MSK, BPSK)	PacketType	
SetModulationParams	0x020F	Configure modulation parameters	Params[7:0]	
SetPacketParams	0x0210	Configure packet parameters	Params[8:0]	
SetTxParams	0x0211	Set Tx power and ramp time	TxPower RampTime	
SetPacketAdrs	0x0212	Set the Node address and the broadcast address for the gfsk/gmsk packets	NodeAddr BroadcastAddr	
SetRxTxFallbackMode	0x0213	Defines into which mode the chip goes after a TX / RX done	Fall-back mode	
SetRxDutyCycle	0x0214	Start Rx Duty Cycle mode	RxPeriod(23:0) SleepPeriod(23:0)	
SetPaConfig	0x0215	Configure PA settings	PaSel RegPaSupply PaDutyCycle PaHpSel	
StopTimeoutOnPreamble	0x0217	Stop Rx time-out on Syncword/Header (default) or preamble detection	StopOnPreamble (0 = stop on Sync/Head, 1 = stop on preamble)	
SetCAD	0x0218	Set chip into RX CAD mode (LoRa)		
SetTxCw	0x0219	Set chip into TX mode with infinite carrier wave		
SetTxInfinitePreamble	0x021A	Set chip into TX mode with infinite preamble		
SetLoRaSynchTimeout	0x021B	Configures LoRa modem to issue a time-out after exactly NbSymbols symbols	NbSymbols Format (optional)	
SetRangingAddr	0x021C	Sets the ranging Id of the device and how many bytes are checked on a request	Addr CheckLength	
SetRangingReqAddr	0x021D	Sets the ranging id for the request	ReqAddr	

Table 5-3: Radio Configuration / Status Commands (Sheet 3 of 3)

Commands	Cmd Id	Description	Input	Output
GetRangingResult	0x021E	Gets ranging results (results, rssi)	Type	Res (31:24) Res (23:16) Res (15:8) Res (7:0)
SetRangingTxRxDelay	0x021F	Sets the TxRx delay of the ranging	Delay	
RfFeTest (LR1110 only)	0x0222	Command to put chip in to several test modes for RF FE tests	TestCase	
SetGfskCrcParams	0x0224	Sets the parameters for the crc polynom	SeedValue, Poly-Value	
SetGfskWhitParams	0x0225	Sets the parameters for the whitening	SeedValue	
SetRxBoosted	0x0227	Sets the Rx to boosted mode or not	Boosted	
SetRangingParameter (LR1110 only)	0x0228	Sets the ranging specific parameters	ExtendedMode, SymbNb	
SetRssiCalibration	0x0229	Sets the RSSI gain tune table	GainTuneTable	
SetLoRaEBlockCfg	0x022A	Configures on hop of LR-FHSS hopping	BlockId, NbSym-bols, Freq	
SetLoRaSyncWord	0x022B	Sets the LoRaSyncWrod	Syncword	
LrFhssBuildFrame	0x022C	Encode the LR-FHSS frame, sets up the hopping table	SyncWordCount, CR, ModType, Grid, Hopping, BW, DeviceOffset, Payload	
LrFhssSetSyncWord	0x022D	Sets the LR-FHSS syncword	Syncword	
ConfigBleBeacon (LR1110 only)	0x022E	Configures radio to be Bluetooth Low Energy beaconing compatible	Channel, Payload	
ReadLoRaEtable	0x022F	Returns the internal LR-FHSS hopping table		
GetLoRaRxHeaderInfos	0x0230	Returns the infos about CR and CRC_en for last received LoRa packet		Infos
BleBeaconSend (LR1110 only)	0x0231	Configures the radio for BLE and sends the given beacon on the asked channel	Channel, Payload	
RadioApplyPatch	0x0234	Allows implementation of workarounds and/or debug features	patch_id, ampDetEn	

5.4 List of Low Latency APIs (LR1121 only)

In order to help to reduce the communication latency between the host MCU and the transceiver, these low latency APIs provide a way to concatenate multiple serial commands in order to reduce the serial protocol overhead. These APIs' results are particularly beneficial when high performances are needed, such as during drone control.

All these commands require Modem-E to be suspended and be updated to version 2.1.0+. Some commands have a 2-phase transaction, a request (Input) and a response (Output).

See the LR1121 Transceiver User Manual for details of these commands.

Table 5-4: Low Latency Commands (Sheet 1 of 2)

Commands	Cmd Id	Description	Input	Output
GetPacket	0x0700	Combines the following commands: GetRxBufferStatus, GetPacketStatus and ReadBuffer8	CmdID	PayloadLen(N) PktStat (31:24) PktStat (23:16) PktStat (15:8) PktStat (7:0) Payload(0)
SetRfFrequency_SetRx	0x0701	Combines the following commands: SetRfFrequency and SetRx	RfFreq (31:24) RfFreq (23:16) RfFreq (15:8) RfFreq (7:0) RxTimeout (23:16) RxTimeout (15:8) RxTimeout (7:0)	-
SetRx_GetPacket	0x0702	Combines the following commands: SetRx and GetPacket	RxTimeout (23:16) RxTimeout (15:8) RxTimeout (7:0)	PayloadLen(N) PktStat (31:24) PktStat (23:16) PktStat (15:8) PktStat (7:0) Payload(0:N-1)
SetRfFrequency_SetRx_GetPacket	0x0703	Combines the following commands: SetRfFrequency, SetRx and GetPacket	RfFreq (31:24) RfFreq (23:16) RfFreq (15:8) RfFreq (7:0) RxTimeout (23:16) RxTimeout (15:8) RxTimeout (7:0)	PayloadLen(N) PktStat (31:24) PktStat (23:16) PktStat (15:8) PktStat (7:0) Payload(0:N-1)

Table 5-4: Low Latency Commands (Sheet 2 of 2)

Commands	Cmd Id	Description	Input	Output
WriteBuffer8_SetTx	0x0704	Combines the following commands: WriteBuffer8 and SetTx	Payload(0:N-1) TxTimeout (23:16) TxTimeout (15:8) TxTimeout (7:0)	-
WriteBuffer8_SetRfFrequency_SetTx	0x0705	Combines the following commands: WriteBuffer8, SetRfFrequency and SetTx	Payload(0:N-1) RfFreq (31:24) RfFreq (23:16) RfFreq (15:8) RfFreq (7:0) RxTimeout (23:16) RxTimeout (15:8) RxTimeout (7:0)	-

6. Revision History

The following table details the versions of the Reference Manual document issued, and the corresponding LoRa Basics™ Modem-E versions supported (Use Case and FW Major.FW Minor), as returned by the command *ModemGetVersion(...)*.

Table 6-1: Revision History

Ref. Manual Version	ECO	Date	Applicable to	Changes
1	-	May 2021	Modem V1	-
2		Sept 2024	Modem V2 Use Case: 02 FW Version: 1.2 or later	First version
3	079031	April 2026	Modem-E V2.1.0+	Added Low Latency commands Added downlink and reset events Removed commands: • WriteMem8 • ReadMem8 • ReadTxBuffer8 • CaptureIq • GetRadioDebugResults • SystemTests • CalibrateRTC • GetRTCres • GetPIIntPath

Glossary

List of Acronyms and their Meaning

Acronym	Meaning
ABP	Activation By Personalization
ADR	LoRaWAN® Adaptive Data Rate
ALC	Application Layer Clock Synchronization (ALC Sync)
BW	LoRa® modulation Bandwidth
CR	LoRa® modulation Coding Rate
DR	Datarate
FOpt	LoRaWAN® Frame Options
FUOTA	Firmware Update Over The Air
LoRa®	Long Range Communication. The LoRa® Mark is a registered trademark of the Semtech Corporation
NVM	Non-Volatile Memory
OTAA	Over-The-Air Activation
RSSI	Received Signal Strength Indicator
RFU	Reserved for Future Use
SF	LoRa® modulation Spreading Factor
TRNG	True Random Number Generator



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